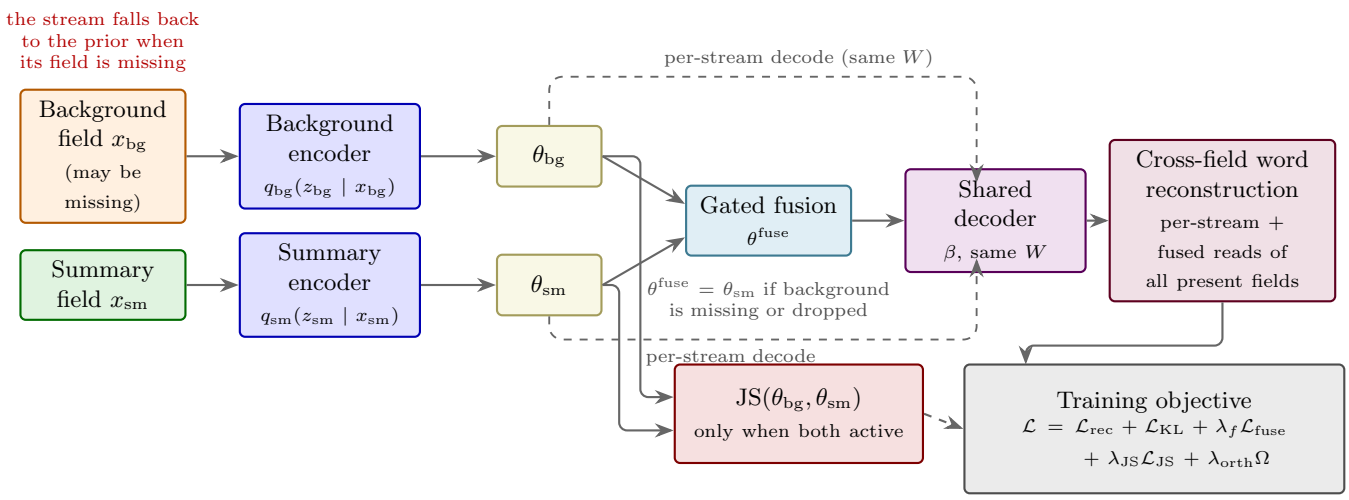

the stream falls back to the prior when its field is missing
Background field x_{bg} (may be missing)
Summary field x_{sm}
Background encoder $q_{\mathrm{bg}}(z_{\mathrm{bg}} \mid x_{\mathrm{bg}})$
Summary encoder $q_{\mathrm{sm}}(z_{\mathrm{sm}} \mid x_{\mathrm{sm}})$
per-stream decode (same W)
θ_{bg}
θ_{sm}
Gated fusion θ^{fuse}
$\theta^{\mathrm{fuse}} = \theta_{\mathrm{sm}}$ if background is missing or dropped
Shared decoder β, same W
Cross-field word reconstruction per-stream + fused reads of all present fields
per-stream decode
$\mathrm{JS}(\theta_{\mathrm{bg}}, \theta_{\mathrm{sm}})$ only when both active
Training objective
$\mathcal{L} = \mathcal{L}_{\mathrm{rec}} + \mathcal{L}_{\mathrm{KL}} + \lambda_f \mathcal{L}_{\mathrm{fuse}} + \lambda_{\mathrm{JS}} \mathcal{L}_{\mathrm{JS}} + \lambda_{\mathrm{orth}} \Omega$